PATRICK KRUSE

Grand Island, NY | 716-249-7385

pekruse504@gmail.com| linkedin.com/in/patrick-kruse-NY | https://pekruse.github.io/patrick-kruse-portfolio

EDUCATION

University at Buffalo - Buffalo, NY *08/2026 - Present*

B.S. in Electrical and Computer Engineering | Law minor | Junior *Expected 2028*

Organizations/Activities: SEDS Avionics | Robotics Club | ACES Tennis Club | Microsoft Learn Student Ambassador

Alfred State College | Alfred University - Alfred, NY *08/2024 - 08/2026*

B.S. in Electrical Engineering Technology | GPA: 3.91

Organizations/Activities: President of Main Gate B Hall Council | Hiking Club | Swimming

Honors: IEEE PES Scholar Recipient (2025-26)

EXPERIENCE

Ark Development Solutions - Private software services startup- Buffalo, NY *10/2026 - Present*

Co-founder

- App development startup that leverages agentic AI and democratized training data to assist disadvantaged groups and address gaps in local services
- Designed a front website hosted with Github pages and IONOS domain name. Current mobile application development and Buffalo community winter outreach

U.S. Air Force | 107th Attack Wing - Niagara Falls, NY *05/2025 - Present*

Cyber Systems Administrator | 107th Attack Wing | National Guard

- Manage networks, devices, user accounts, and infrastructure security for highly critical and classified MQ-9 Reaper mission
- Run server rack cables, perform system configuration and patch management. Provide service to users experiencing issues.

Alfred State College of Technology - Alfred, NY *01/2025 - 12/2025*

Peer Tutor

- Tutored 30+ students in math, programming, and engineering courses
- Writing & Math Lab Proctor

PROJECTS

- (Current) Semi-autonomous VTOL drone propelled with four tilt-rotor vector control and 1m wingspan. Airframe aerodynamics and electrical systems modeled in Simulink. Plan to add precision magnetometer for secondary coordinate mapping, camera, and laser turrets for real-time threat detection and response.
- Sensor-based Handwriting Recognition Device
- Motion-triggered water dispenser with touchless overfill protection for bottles of various sizes and materials. Combines real-time microphone data, analog filtering, digital signal processing, and acoustic principles to pinpoint bottle geometry and remaining volume.
- Class AB Audio Amplifier achieving 35% efficiency. Custom-built AC-DC linear regulated power supply.

SKILLS & CERTIFICATIONS

Equipment: Multimeter, Oscilloscope, Function generator, Spectrum analyzer,
Hardware: Arduino, Raspberry Pi, ESP32, STM32, Cyclone V FPGA, Click PLC
Software: LTSpice, Altium Designer, KiCad, Ansys Electrical, Intel Quartus, CODESYS, Factory IO, SolidWorks, CISCO PT
Programming: MATLAB/Simulink, Python, C#, C/C++, Verilog
Data Analytics & AI Automation: Microsoft Excel, PowerBI, Jupyter Notebook, PyTorch, Claude, Codex
Certifications/Licenses: CompTIA Security+, Active TS/SCI security clearance, Lean Six Sigma Yellow Belt

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- 3-drone mesh network utilizing magnetometer and electromagnet near-peer tracking system and cognitive spectrum defense to resist wireless jamming
- Day trading/arbitrage FPGA bot achieving under 100ns of latency and 55% signal-to-execution accuracy
- Two-stage audio amplifier with custom 12V AC-DC linear regulated power supply, speaker, and audio jack
- 2’x2’x3’ wooden trebuchet with metal disc counterweight that launched lacrosse ball over 40 feet. Leveraged MATLAB to simulate rigid-body mechanics and flight trajectory

SKILLS AND PROJECTS

Hands-on skills

- Soldering, troubleshooting, schematic & PCB design
- Experience with multimeters, oscilloscopes, power supplies, signal generators, spectrum analyzers
- Woodworking, sheet metal, machine tools

Engineering tools

- LTSpice, Altium Designer, KiCad
- Intel Quartus, AMD Vivado
- Industrial automation simulation with CODESYS, Factory IO
- MATLAB, Jupyter Notebook
- CISCO Packet Tracer

Programming & IT skills

- PLC & FPGA programming
- VHDL, Verilog
- C, C#, Python, Arduino IDE
- Linux/PowerShell scripting
- Network/IoT architecture modeling

Administrative skills

- Project design and management
- Microsoft Excel, Word, PowerPoint, Google suite
- Can use AI to automate tasks with MCP/API connectors

Certifications and Credentials

- Active DoD TS/SCI clearance
- CompTIA Security+ certified
- CompTIA CySA+ certified
- CCNA certified
- FE Exam Electrical and Computer []
- FCC Amateur Radio License: Amateur Extra Class - Call Sign: []
- SolidWorks CWSP
- Lean Six Sigma Green Belt
- OSHA 10
- CPR/EMT-B licensed (fire dept.)

Projects (limit to 3-4 great but diverse projects)

- Motion-triggered water dispenser with touchless overfill protection for bottles of various sizes and materials. Combines real-time microphone data, DSP filtering, and acoustic principles to pinpoint bottle geometry and remaining volume
- 3-drone mesh network utilizing magnetometer and electromagnet near-peer tracking system and cognitive spectrum defense to resist wireless jamming
- Low-power, offline cryptographic door lock system integrating ESP32 controllers, IR/BLE handshakes, and hardware security features like optocoupler isolation
- Hardened phone screen overlay handset that utilizes HDMI screen sharing and HID commands to emulate user interaction on actual mobile device, locally encrypt chat and email communications, store cryptocurrency secrets, and sign transactions offline. Resistant to state-sponsored mobile spyware such as graphite.
- Day trading/arbitrage FPGA bot achieving under 100ns of latency and 55% signal-to-execution accuracy
- Two-stage audio amplifier with custom 12V AC-DC linear regulated power supply, speaker, and audio jack
- 2’x2’x3’ wooden trebuchet with metal disc counterweight that launched lacrosse ball over 40 feet. Leveraged MATLAB to simulate rigid-body mechanics and flight trajectory

- UB Students for Exploration and Discovery of Space (SEDS) Club - avionics hardware team
- UB Robotics Club - participate in the design & construction of a four-legged, flying/walking drone
- ACES Tennis Club — team practices and competitive matches
- Hall Council President — planned and organized funded residence hall events with food, games, and music; led team of five members
- IEEE PES Scholarship Plus Scholar Recipient (2025–26)

EXPERIENCE & ORGANIZATIONS

CampusDash Cofounder

- **Designed and shipped a peer-to-peer food delivery service app serving 150+ orders for the UB student population daily**
- **Android and iOS app versions seamlessly utilize secure university-sanctioned CBORD GET email receipt notifications to track order times and calculate delivery fees**
- **Created 50+ private student campus jobs and generated over $3000 in net profit in the first month of operation**

[EXAMPLE] University at Buffalo — Electrical Engineering Department *11/2026 – Present*
Undergraduate Researcher

- Assisted research in optically accelerated hardware architectures for integration of homomorphic encryption into legacy processing

New York Air National Guard, 107th Attack Wing — Niagara Falls, NY *05/2025 – Present*
Cyber Systems Administrator

- Manage networks, organization devices, user accounts, and infrastructure security for MQ-9 Reaper mission
- Run cables, perform system configuration and patch management
- Provide technical service to users in the Group experiencing issues

Tops Markets — Grand Island, NY *05/2025 – 02/2026*
Bakery Associate

- Managed bakery display tables, cases, and freezer inventory
- Handled dozens of customer requests, take/fulfill orders, and pipe inscriptions/borders on cakes

SUNY Alfred State College of Technology — Alfred, NY *01/2025 – 12/2025*
Peer Tutor

- Tutored 20+ students for math, programming, and engineering courses; actively kept recurring clients; Writing/Math Lab proctor

Environmental Services Group (now Momentum Environmental) — Tonawanda, NY *07/2023 – 09/2023*
Administrative Clerk

- Organized hundreds of files, performed data entry and spreadsheet design, maintained office

VOLUNTEER WORK

Grand Island Fire Company — Grand Island, NY *02/2024 – Present*
Senior Cadet

- Practice basic first response and operational equipment skills through weekly drills and EMS calls